

An End-to-End Framework for Electrical Metrology Digitalization

API-based DCC Integration, XSLT Visualization, and Workflow Automation
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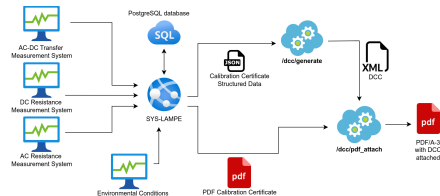
Outline

- 1 Work Focus
- 2 Framework Architecture
- 3 Human-Readable Delivery
- 4 Pilot and Results

Scope of this contribution (for a DCC-specialized audience)

This presentation focuses on implementation details and practical results, assuming prior familiarity with DCC concepts. **What is new since our previous report:**

- Consolidated Flask-based service architecture (Docker deployment)
- Expanded data model for multiple quantities and multi-index tables
- XSLT-based DCC visualization workflow integrated in the web app
- PDF/A-3 attachment pipeline with SHA256 integrity artifact
- Pilot-grade end-to-end integration in Inmetro laboratories (LAMPE and LACEL)



Technical contributions in this version

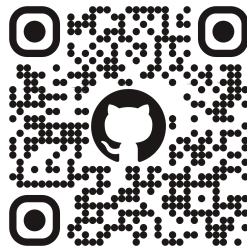
Main objective

Deliver an **operational end-to-end framework** for generation, visualization, validation, delivery, and downstream use of DCCs.

- Route /dcc/generate: JSON to DCC XML (schemaVersion 3.2.0)
- Route /dcc/upload_xls: Excel workbook to JSON to DCC XML
- Route /dcc/visualizar_dcc: XML to HTML via XSLT template
- Route /dcc/pdf_attach: XML embedding in PDF/A-3 plus SHA256SUM.txt
- Route /dcc/validate_xml: XML schema validation with automatic XSD import/include loading

Implementation architecture and deployed stack

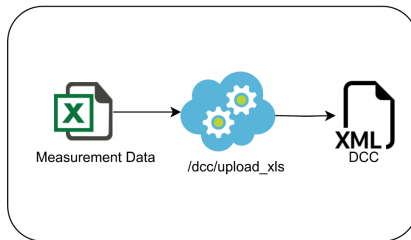
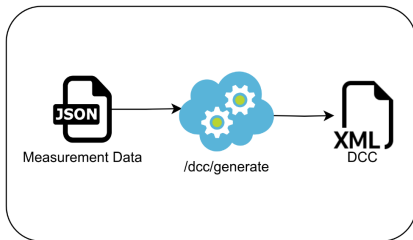
- Open-source implementation:
https://github.com/gmgeronymo/dcc_tools
- Current production-oriented stack: Python Flask microservice
- Deployment model: Docker Compose (default service on port 9099)
- Core libraries: lxml, pandas/openpyxl, pikepdf
- Data input routes:
 - JSON for system-to-system automation
 - Excel for operator-assisted laboratory workflows
- Output: PTB DCC XML + optional visualization/validation/attachment services



Certificado de Calibração Digital (DCC)

Publicações	FAQ (perguntas frequentes)	Exemplos
Excel para DCC	JSON para DCC	Embutir DCC no PDF
Validar XML	Documentação da API	Visualizar DCC
GEMMIMEG Tool (V1)	Código-Fonte (GitHub)	

Implemented API workflow and endpoints



/dcc/generate, /dcc/upload_xls, /dcc/pdf_attach, /dcc/visualizar_dcc, /dcc/validate_xml

Gerar DCC a partir de arquivo JSON

[Exemplo de arquivo JSON](#)

Arquivo JSON:

Choose File No file chosen

Enviar

Gerar DCC a partir de planilha Excel

[Exemplo de planilha Excel](#)

Planilha Excel:

Choose File No file chosen

Enviar

Simplified JSON model: implementation decisions

- Administrative blocks mirror Inmetro certificate metadata fields
- `mensurando` supports multiple quantities with independent units and labels
- `indices` supports variable-dimensional indexing per quantity (range, voltage, frequency, ...)
- `resultados` stores value/uncertainty/coverage factor with quantity linkage
- Relative uncertainty mode (ppm) is converted to absolute uncertainty in generation stage

Input JSON example used by the API

```
{
  "mensurando": [
    {"label": "acdc", "name": "AC-DC transfer difference",
     "col_name": "delta_u", "unit": "\\micro\\volt\\volt\\tothe{-1}",
     "unc_relativa": false}
  ],
  "indices": [
    {"mensurando": "acdc", "label": "range", "unit": "\\volt"},
    {"mensurando": "acdc", "label": "frequency", "unit": "\\kilo\\hertz"}
  ],
  "resultados": [
    {"mensurando": "acdc", "range": "0.022", "frequency": "0.01",
     "value": "-280", "unc": "44", "k": "2.13"}
  ]
}
```

This structure is shared by /dcc/generate and /dcc/upload_xls.

Corresponding generated DCC XML (excerpt)

```
<dcc:result>
  <dcc:name><dcc:content lang="pt">AC-DC transfer difference</dcc:content></dcc:name>
  <dcc:data>
    <dcc:list>
      <dcc:quantity>...<si:valueXMLList>0.022</si:valueXMLList>
        <si:unitXMLList>\volt</si:unitXMLList></dcc:quantity>
      <dcc:quantity>...<si:valueXMLList>-280</si:valueXMLList>
        <si:unitXMLList>\micro\volt\volt\tothe{-1}</si:unitXMLList>
        <si:uncertaintyXMLList>44</si:uncertaintyXMLList>
        <si:coverageFactorXMLList>2.13</si:coverageFactorXMLList>
      </dcc:quantity>
    </dcc:list>
  </dcc:data>
</dcc:result>
```

Generated by /dcc/generate from the previous JSON structure.

XSLT visualization pipeline (new module)

- Implemented at route /dcc/visualizar_dcc
- Transformation engine uses lxml.etree.XSLT
- Current template path:
static/examples/xslt/hr_template.xsl
- Output is HTML rendered from the same DCC XML payload
- Improves verification without duplicating certificate data sources



XSLT rendering examples

```
<si:unit>\percent</si:unit>
<si:expandedUnc>
  <si:uncertainty>2.8</si:uncertainty>
  <si:coverageFactor>2</si:coverageFactor>
  <si:coverageProbability>0.9545</si:coverageProbability>
</si:expandedUnc>
</si:real>
</dcc:quantity>
</dcc:data>
</dcc:influenceCondition>
</dcc:influenceConditions>
<dcc:results>
  <dcc:result>
    <dcc:name>
      <dcc:content lang="pt">Resistência em Corrente Contínua</dcc:content>
    </dcc:name>
    <dcc:data>
      <dcc:list>
        <dcc:quantity>
          <dcc:name>
            <dcc:content lang="pt">Tensão</dcc:content>
          </dcc:name>
          <si:realListXMLList>
            <si:valueXMLList>10</si:valueXMLList>
            <si:unitXMLList>\volt</si:unitXMLList>
          </si:realListXMLList>
        </dcc:quantity>
        <dcc:quantity>
          <dcc:name>
            <dcc:content lang="pt">V.m.</dcc:content>
          </dcc:name>
          <si:realListXMLList>
            <si:valueXMLList>100.000134</si:valueXMLList>
            <si:unitXMLList>\kilo\ohm</si:unitXMLList>
            <si:expandedUncXMLList>
              <si:uncertaintyXMLList>3.90E-05</si:uncertaintyXMLList>
              <si:coverageFactorXMLList>2.00</si:coverageFactorXMLList>
              <si:coverageProbabilityXMLList>0.9545</si:coverageProbabilityXMLList>
```



Certificado de Calibração

DIMCI 0496/2025
Número de Certificado

Características do Item

Resistência Nominal: 100 kΩ

Rastreabilidade

Os resultados da calibração são rastreados ao Sistema Internacional de Unidades (SI), por intermédio dos padrões metroológicos nacionais. As medições realizadas estão referenciadas ao padrão primário de resistência elétrica do Inmetro, baseado na constante de von Klitzing (efeito Hall quântico) e estabelecido através das comparações chave BIPM.EM-K13.a e BIPM.EM-K13.b de 2022. As medições realizadas estão referenciadas aos padrões relacionados.

TABELA 1 - Padrões Utilizados			
Descrição	Identificação	Certificado	
		Nº/Ano	Origem
Resistor Padrão	Lampe PR 405	DIMCI 1263/2024	Inmetro

Método de Medição

O resultado fornecido refere-se ao valor médio de três séries de seis medições pelo método de comparação potenciométrico por divisor de tensão binário na configuração de 04 (quatro) terminais. Utilizou-se uma ponte automática de resistência modelo 6000A.

Informações Pertinentes às Atividades Realizadas

Condições Ambientais:

Temperatura no banho de ar: (23,06 ± 0,07) °C

Umidade relativa: (54,4 ± 2,8) % ur

Resultados e Declaração da Incerteza de Medição

As incertezas expandidas de medição (U) relatadas são declaradas como a incerteza padrão combinada multiplicada pelo fator de abrangência k, que, para uma distribuição t, com um número efetivo de graus de liberdade v_{eff}, o qual corresponde a uma probabilidade de abrangência de 95,45 %. A incerteza de medição expandida foi relatada de acordo com a publicação Avaliação de Dados de Medição - Guia para a Expressão de Incerteza de medição - GUM 2008.

TABELA 2 - Resistência em Corrente Contínua			
Tensão (V)	V.m. (kΩ)	U	k
10	100,000134	3,90E-05	2,00

PDF/A-3 attachment module: implementation details

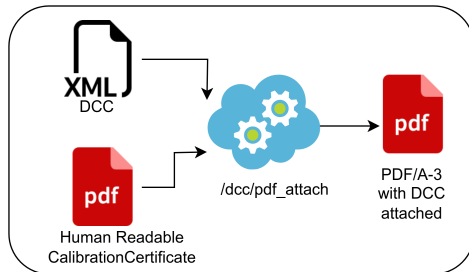
Incluir DCC no PDF do Certificado de Calibração

Arquivo XML:

No file chosen

Arquivo PDF:

No file chosen



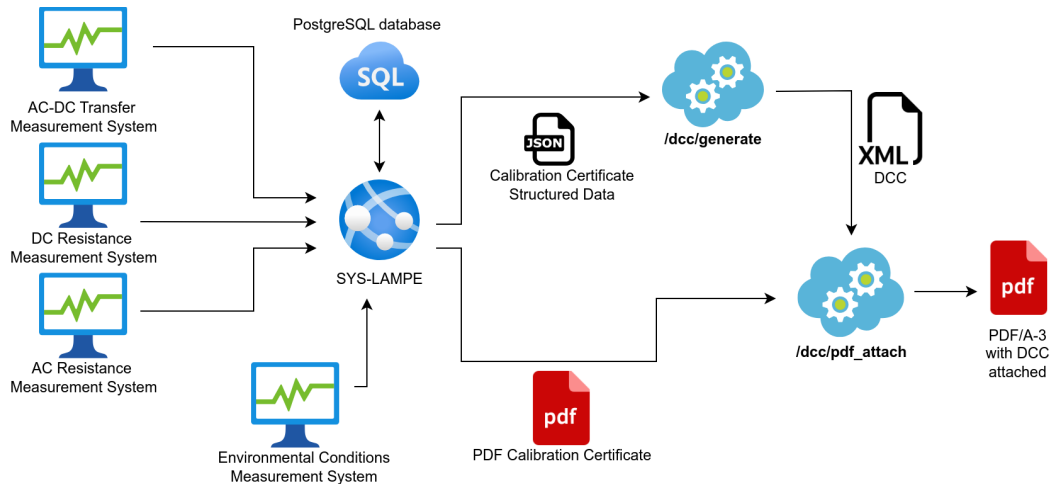
- Implemented at route `/dcc/pdf_attach`
- XML attached as EmbeddedFile with `AFRelationship=Alternative`
- Names tree updated for attachment navigation in common PDF readers
- Additional attached file: `SHA256SUM.txt` for integrity verification

Attachment accessibility in standard PDF readers



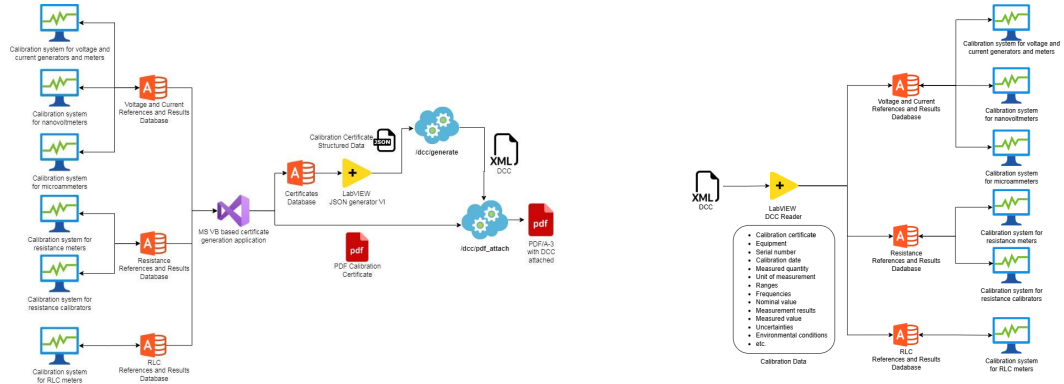
Operational behavior verified in common readers while preserving machine extraction of XML

Pilot deployment: LAMPE and LACEL laboratories



SYS-LAMPE / LACEL databases → JSON → API → DCC XML → PDF/A-3 with attachment

Automated DCC processing in LabVIEW



DCC Reader VI extracts calibration data and updates reference databases automatically.

Observed impact and open technical points

Operational gains in the pilot

- Elimination of manual data transcription from certificates
- Reduced risk of transcription errors in high-point calibrations
- Faster update cycle for reference data in automation systems
- Stable interoperability between generation and LabVIEW processing workflow

Open points under active development

- Harmonized refType strategy and ontology/PID alignment
- Progressive adoption of multilingual fields and richer semantic tagging
- Expansion of validated integrations across additional laboratory stacks

Thank you for your attention

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